

Lily Oglesby

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EDUCATION

University of Southern California <i>Master of Business Research, Marshall School of Business</i>	Los Angeles, CA Aug 2025 – Dec 2026
University of California, Berkeley <i>Bachelor of Arts in Astrophysics and Data Science</i>	Berkeley, CA Aug 2021 – May 2025

RESEARCH EXPERIENCE

Space Sciences Laboratory, UC Berkeley <i>Undergraduate Researcher</i>	Berkeley, CA Aug 2021 – May 2025
◦ Satellite Data Analysis: Performed time-series preprocessing, statistical filtering, and exploratory analysis on three years of ICON satellite data to identify ionospheric signatures of geomagnetic storms. ◦ Scientific Programming: Developed reusable Python workflows to filter, analyze, visualize, and communicate satellite-data results.	
Institute for Astronomy, University of Hawaii at Manoa <i>Research Intern</i>	Honolulu, HI Jun 2024 – Aug 2024
◦ Machine Learning: Applied unsupervised anomaly detection to approximately 120,000 telescope images and used a convolutional neural network to classify galaxy morphology. ◦ Computational Efficiency: Used GPU parallelization to accelerate large-scale astronomical image processing and model execution.	

TECHNICAL PROJECT

Interactive Pokémon-Style Portfolio Website <i>Developer and Designer</i>	2026
◦ Web and Game Development: Built an interactive portfolio using JavaScript, HTML, CSS, Canvas, GSAP, Howler.js, and Tiled, featuring an explorable map, mobile controls, animation, audio, and turn-based battles. ◦ Programming and Deployment: Developed and debugged movement, collision detection, battle mechanics, asset loading, audio, and responsive behavior; managed the project with Git and GitHub and deployed it through Netlify.	

TEACHING AND MENTORING EXPERIENCE

USC DSO Research Experiences for Undergraduates Program <i>Graduate Student Instructor and Mentor</i>	Los Angeles, CA Jun 2026 – Present
◦ Research Instruction and Advising: Led sessions on responsible uses of artificial intelligence in research and advised students on graduate applications, research statements, and presenting technical experience.	
Introduction to General Astronomy, UC Berkeley <i>Reader and Grader</i>	Berkeley, CA Aug 2022 – Dec 2024
◦ Course Assessment: Evaluated assignments across multiple semesters and provided feedback on foundational physics and astronomy concepts.	
Mauna Kea Scholars Program, University of Hawaii <i>Research Mentor</i>	Honolulu, HI Jun 2024 – Aug 2024
◦ Research Mentorship: Mentored a high school student in astronomy research and taught Linux, telescope-data analysis, ASTRONOMY, and research presentation skills.	
Python Programming in Astronomy, UC Berkeley <i>Reader, Grader, and Teaching Assistant</i>	Berkeley, CA Jun 2023 – Aug 2023
◦ Programming Instruction: Supported students through office hours and evaluated Python-based astronomy assignments for correctness and clarity.	

OUTREACH AND LEADERSHIP

Royal Astronomical Society of Hawaii <i>Volunteer Outreach Coordinator</i>	Honolulu, HI Jun 2024 – Aug 2024
◦ Community Engagement: Organized public stargazing events and led astronomy education incorporating Hawaiian perspectives on the night sky.	
NASA Astrobiology Outreach, UC Riverside <i>Project Leader</i>	Riverside, CA Aug 2017 – Jan 2020
◦ STEM Education: Led hands-on astrobiology programs for elementary students and public events and trained outreach volunteers.	

SKILLS SUMMARY

Programming: Python, JavaScript, HTML, CSS, Git, GitHub, Linux, LaTeX
Data Science: Machine Learning, Statistical Analysis, CNNs, Anomaly Detection, Data Visualization, Image Processing
Development: Canvas API, GSAP, Howler.js, Tiled, Responsive Design, Netlify